Product 2: Vendor-Agnostic Inverter and Battery Controller

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Summary

Build Product 2, but sell it with a published list of 20 to 30 supported device families, not as "works with all systems". Every mature product grows coverage one tested template at a time, and vendor firmware keeps changing the maps (evcc templates; deye-inverter-mqtt).

Net billing credits new exports at Rs 8 to 11/kWh while imports cost Rs 37 to 55/kWh (Dawn; NEPRA SRO 1501). Battery and export control therefore pays. No local offering found publishes a device list for Chinese hybrids and LFP packs, or a battery state independent of the OEM BMS.

Finding	Number	What it means for Product 2
Every project identifies a device, then loads a hand-written template; none infers the map	evcc 251 meter templates, 134 brands; ha-solarman 30 profiles; SolaX plugin 126 to 130 serial-prefix branches (evcc; solax-modbus)	"Detect all patterns" means a curated library plus automatic identification
Automatic inference of register meaning is weak	Best Modbus field inference F1 about 0.70; building point typing 82 to 92%, 99% only after 100 to 160 labels (DynPRE; Scrabble)	ML drafts a map for an installer to confirm; it cannot run unattended
Firmware routinely changes maps	Deye HV firmware 1098+ changed energy scaling x10; Sungrow ships 1 to 3 updates a year without changelogs (Deye; Sungrow FAQ)	Templates keyed to firmware, with run-time sanity checks
Cheap BMS state of charge (SoC) is unreliable on LFP	1 mV voltage error is about 5% SoC; a JK BMS read 85% when the pack was empty (LFP paper; Victron forum)	An independent shunt (from US$100.30) plus an estimator is the headline feature

Finding	Number	What it means for Product 2
Local rivals already sell OEM-agnostic gateways	NeoTAQ claims 50+ Pakistani sites and 100 MW+; its sole distributor GFE also sells Elum (NeoTAQ; GFE)	Win on Chinese hybrid coverage, battery trust and price, or partner
Careless writes damage devices	About 100,000 EEPROM cycles (community); SMA warns cyclic changes destroy flash (SolaX docs; SMA)	Write only whitelisted volatile registers, with rate limits and revert timers

Is "works with all systems" achievable? Not literally. It is achievable as "works with every listed device, and onboards an unlisted one in under an hour with the installer". All surveyed projects rely on curated templates plus identity lookup (Victron probe.py; evcc detect).

What "detect all patterns and adapt" realistically means. Four mechanisms, most reliable first:

1. Curated templates: one data file per device family, with firmware variants and a recorded test dump.
2. Deterministic identification: read the SunSpec marker, Modbus device ID or a vendor model register, then pick the template.
3. Plausibility scoring: decode candidate maps read-only, score how sensible values look, and flag silent firmware changes.
4. ML-assisted mapping in an installer tool: capture vendor traffic, run physics checks, compare with a reference meter, then get installer confirmation.

Hardware box or software on gateways? Ship the software on an OnyxLeaf box built from an off-the-shelf Linux module (Raspberry Pi CM5 class). It needs two isolated RS485 ports, one isolated CAN port and a shunt input. A 4G router would push safety logic into the cloud, and an ESP32 is too small.

Defer a custom board until 20 to 50 pilot sites have run. Containerise the software so partners' Linux gateways can host it later.

Refined problem statement

The research narrows "any inverter, battery and meter" to 5 inverter families, 2 meter types and 2 battery protocols for release 1. Everything else goes through guided onboarding.

What the meeting notes asked, and what changes

Meeting note	Research finding	Revised wording
"Works with all systems"	Coverage grows template by template	"Works with every listed device; guided onboarding for the rest"
"Detect all patterns and adapt"	Identify-then-template everywhere	"Identifies listed devices; maps unknown ones with installer confirmation"
"Sequencing of serial data is specific"	Three things: transport framing, word order per point, block and timing limits per device	All three become fields in a device data file
"OEM supplier shows wrong data of battery"	LFP physics plus decoding errors	"Independent battery state with a trust score"
"Buy inverter of my company"	OEM tools work best inside one brand, e.g. Sungrow EMS300CP	"One controller and one data model across brands"
"Controller design OR frontend and backend"	The device layer is defensible; dashboards are crowded	Edge box and installer tool first; thin fleet backend

Precise problem statement

Installers in Pakistan build solar-plus-storage sites that mix Chinese inverters (often rebadged), LFP packs and meters. Each vendor exposes data through its own register map, dongle and cloud. No single vendor's tools see a mixed site whole, which pushes owners to "buy our inverter". The OEM BMS also often misreports state of charge.

So no one runs one control policy across a site, export limits are set by hand, and dispatch rests on untrusted battery data. Product 2 reads and safely commands listed devices through one box, normalises them into one data model, and publishes battery state with a confidence. Product 1 (the dispatch optimiser) reads that data and sends setpoints back; Product 2 enforces them safely.

Scope

Area	In scope, release 1	Out of scope, release 1
Devices	Hybrid and string inverters, LV LFP BMS, grid and load meters	EV chargers, HVAC, utility plant controllers
Protocols	Modbus RTU and TCP, Solarman V5, Voltronic ASCII, Pylontech-style CAN, Pylon RS485	IEC 61850, DNP3, IEEE 2030.5 (no local rule found requires them)
Control	Work mode, battery power or current, export limit, TOU slots, via whitelisted registers	Grid-protection settings, third-party firmware updates, writes to unknown registers
Battery	Independent SoC and state of health (SoH); trust score	Replacing or reflashing the BMS
Gensets	Read status over Modbus	Start, stop and synchronisation (Elum, ComAp and Enercon own this)
Optimisation	Local API for Product 1 setpoints	Forecasting and optimisation
Unknown devices	Installer onboarding tool with confirmation	Zero-touch automatic mapping
Cloud	Outbound sync, fleet health, installer accounts	Full analytics dashboards
Markets	Pakistan pilots; Saudi Arabia after SABER and CST approval	PV mounting structures (on hold)

First target segment

Start with commercial and small industrial sites of about 20 kW to 500 kW with PV, an LFP battery and a TOU or net-billing tariff. The size band is a hypothesis for the device census (WP1).

Industrial B-2 TOU runs Rs 35.68 peak against Rs 22.83 off-peak (IESCO). Diesel genset energy costs about Rs 110 to 142/kWh at Rs 400.67/L (Daily Pakistan; Enpower). The large-industrial share of lithium imports rose from 3.2% in 2025 to 13.9% in January to May 2026 (PSA).

Homes come later: a home monitor like SolarAssistant sells for USD 59 to 229 one-off (SolarAssistant shop). Towers come later: Huawei and Jazz already run an integrated package at 1,000 sites (DCD). Saudi tariffs are flat and surplus earns 5 halalas/kWh, so

Saudi value sits at diesel-heavy sites (Haala Energy).

First brands

Family	Why first	Identification hook
Sungrow SH hybrid and SG string	Named in the meeting; official map public; has an "external EMS" mode	Device type code, register 5000
Growatt SPH, SPA, MIN/MOD TL-XH, then SPF	Named in the meeting; official 85-page map public	Model at holding 28
Deye platform (Sunsynk and other rebadges)	Deepest open-source coverage; many rebadges	Device type, holding register 0
Huawei SUN2000 and LUNA2000	Most frequent brand on named Pakistani C&I sites	Modbus device ID (FC 43/14); model at 30000
Voltronic PI30-family off-grid units	Large off-grid base; open ASCII protocol	"QPI" returns the protocol ID
Meters: Eastron SDM630, Chint DTSU666	Common; public maps	Model registers
Batteries: Pylontech-style CAN, Pylon RS485; then JK, Seplos	De facto LV protocols	Listen for CAN IDs 0x351 to 0x35E

Second wave: GoodWe (Nizam, Premier Energy), Solis and Sofar (Fronus shows INVT and SofarSolar certificates), SolaX and SMA (Nizam; Fronus). Fingerprint Inverex, Fronus and Knox units on site; their OEM platforms are unverified.

Technical background

Most devices on Pakistani sites speak Modbus RTU over 2-wire RS485 at 9600 baud, 8N1 (solax-modbus baud rates). Voltronic-type off-grid units use a 2400-baud ASCII protocol (PI30). The hard parts are bus ownership, register conventions and firmware drift.

Physical layer and bus access

RS485 Modbus allows one master per bus. A vendor dongle, or the inverter's own meter or BMS link, often blocks a second reader.

- Huawei SDongleA-05 and SDongleA-FE block RS485 access while fitted (huawei_solar wiki).

- Sungrow WiNet-S serves one device and fewer registers than the internal LAN port (datasheet; mkaiser).
- Newer Deye units share one RJ45 between RS485 (pins 1 to 3) and battery CAN (pins 4 and 5) (kellerza wiring).
- A DTSU666 on a Sungrow or Huawei meter port must not get a second master (evcc).

Workarounds, in order of preference:

1. Use a second port, such as Huawei RS485A1 or the Sungrow internal LAN.
2. Read meter and battery values through the inverter that masters them.
3. Put one proxy on the bus: mbusd turns an RTU bus into one TCP server (kellerza mbusd).
4. Use the dongle's local protocol: Solarman V5 on TCP 8899 carries Modbus without disturbing the cloud (pysolarmanv5).
5. Proxy the Growatt cloud link, as grott does via datalogger registers 17 and 18 (grott).
6. Sniff passively with a high-impedance listener, or replace the dongle.

Use FT232RL USB adaptors and Modbus-aware Ethernet gateways; transparent adaptors cannot route to the right slave (adaptor list).

Modbus quirks that break generalisation

The same quantity differs in 12 ways across vendors, and sometimes across firmware of one vendor. Each must be a per-point or per-device field.

Quirk	Effect	Vendor examples
Address base	Off-by-one reads	Sungrow documents are 1-based (Sungrow V1.1.9)
Function code	FC03 and FC04 hold different data	Solis runtime FC04 33xxx, settings FC03 43xxx; SDM630 FC04 vs DTSU666 FC03 (mbmd)
32-bit word order	Silent scrambling	Sungrow U32 low word first; SolaX defaults to little
Sign and scale	Plausible wrong numbers	Deye signed power, x0.1 points; one SolaX model off by x10 (SolaX FAQ)
Sentinels	0xFFFF read as 6553.5	Sungrow 0xFFFF and 0x7FFF
Timing	Timeouts, lockups	Sungrow 1,000 ms; Growatt 850 ms (V1.24); Huawei 10 s timeout (lib)

Quirk	Effect	Vendor examples
Registers per read	Failed reads	Growatt 125; Huawei 64; SH5.0RS on WiNet-S 38 (SunGather)
Atomic writes	Single writes rejected	Growatt TL-XH 3038 and 3039 in one FC16 (evcc)
Slave ID	No answer	Huawei 0 or 1; GoodWe 0xF7; Deye resets to 0 after some upgrades (kellerza)
Packed fields	Wrong times and models	Growatt hour and minute in one word; FoxESS two ASCII characters per register
Write unlock	Writes ignored	SolaX "Lock State"; Huawei installer login
Persistence	EEPROM wear	SolaX 0x612 EEPROM vs 0x669 RAM; Growatt 30215 vs volatile 30216 (plugin_growatt)

Standards

- SunSpec Modbus is the only open, self-describing map: "SunS" at 40000, 50000 or 0, then models 1, 101/103, 124, 701 to 715 and 802 (sunspec/models). Western brands use it; evcc's Growatt and GoodWe templates are "manual non-sunspec" (evcc).
- SunSpec carries safe-control primitives: reversion timers in models 704 and 124, a controller heartbeat in 802 (model 704).
- IEEE 1547-2018 requires a local interface via IEEE 2030.5, IEEE 1815 or SunSpec Modbus (NRECA slides). IEEE 1547.1-2020 defines 47 test areas; CSIP builds on 2030.5 (QualityLogic).
- Pylontech-style LV CAN runs at 500 kbps: 0x351 limits, 0x355 SoC and SoH, 0x356 voltage, current and temperature, 0x359 alarms (CAN V1.3). Most batteries on Deye's list use CAN; about 10 use RS485 (Sunsynk list).
- Voltronic PI30 commands include QPI, QPIGS, POP (output priority), PCP (charger priority) and PSAVE (pi30.py). IEC 61850-7-420 was not verified.

Per-vendor register maps

Register numbers follow each vendor's document; treat them as pointers for template authors.

Vendor	Protocol	Map availability	Key reads	Key writes	Quirks
Sungrow SH hybrid	RTU 9600, ID 1; TCP 502 on LAN	V1.1.9, marked not for redistribution	5000 type; 5214-15 battery W; 5601 meter W; 13008 load	13050 EMS mode (3 = external); 13051 command; 13052 power; 13074 and 13087-88 feed-in limit	1-based; 1 s period
Sungrow SG string	As above	V1.1.53	5000 type	5006 start/stop; 5007-08 power limit; 5010-11 export limit	Export limit on some models only
Growatt	RTU 9600 or 38400	V1.24	Windows differ by family	122-123 export limit; 1044 priority; 1090-92 charge; 1100-08 slots	Packed times; FC16 groups
Huawei	TCP 502 or 6607; RTU RS485A1	huawei-solar-lib (AGPL)	30000 model; 32080 W	47247/47249 forced power; 47416 feed-in limit	Installer login; slow dongle
Deye and rebadges	RTU on RJ45; V5 on TCP 8899	V118 cited in ha-solarman	0 type; 183 battery V; 190 battery W	243-248 priority, load limit, max sell; 250-255 slots (sunsynk)	3-phase uses other addresses
Solis	RTU; V5	Ver3.2 cited in solis_modbus	35000 or 3041 model	43110 storage switch; volatile dispatch	Dispatch needs firmware flag
GoodWe	UDP 8899 AA55; TCP 502	goodwe library	Discovery reply	47509-10 export limit; 47511 EMS mode	Firmware-gated blocks
SolaX	RTU 9600 to 115200	solax-modbus	Serial prefix	Remote control at 0x7C (volatile)	Unlock first
Victron GX	TCP	attributes.csv	843 SoC	2700 AC setpoint; 2902 mode	Unit ID 100
Voltronic PI30	ASCII 2400	PI30 PDF	QPIGS, QMOD	POP, PCP, MCHGC	PSAVE writes EEPROM

Meter or battery	Protocol	Key points	Quirk
Eastron SDM630	RTU FC04 floats	Total power 0x0034 (mbmd)	Variants differ
Chint DTSU666	RTU FC03 floats	Power 0x2012; Sungrow variant ID 254, type 0x20D5 (Sungrow DTSU666)	Firmware variants per inverter brand

Meter or battery	Protocol	Key points	Quirk
Pylontech-style CAN	500 kbps	0x351 to 0x35E; inverter answers 0x305	Groups offset 0x1000
Pylon RS485	ASCII	V3.3 spec in python-pylontech	Differs from CAN path

Pakistan brand mix and local players

No brand-level market share data was found; what is known comes from distributor line-ups, retail listings and about 10 named C&I sites. A device census through installers is therefore the first work package.

What is known and not known

- Distributors: Premier Energy carries GoodWe, Huawei, Infini and Tranergy; Nizam carries GoodWe, SAJ and TBB (Premier; Nizam).
- Named C&I sites run Huawei, SMA and Sungrow; GIKI runs 20 multi-brand inverters (GFE; NeoTAQ case).
- Retail LFP medians (Rs/kWh, Oct 2026): Sunwoda 44,800, Inverex 55,900, Dyness 56,000, Pylontech 67,200 (SolarCitizen).
- Scale: off-grid solar exceeds 13,000 MW (Dawn); 2025 lithium imports were 4.6 GWh, or 3.31 GWh by customs data (ESS News).
- Unknown: brand shares by segment, and the OEM behind Inverex, Knox, Itel and Tesla Industries units.

Rebadges and OEMs

- Fronus shows OEM certificates from INVT and SofarSolar, and signed a 100 MW deal with SolaX in 2022 (Fronus; SolaX).
- Inverex sells NitroX hybrids and its own 51.2 V packs (Inverex). A "-5G" NitroX suffix matches Solis naming; unverified.
- Voltronic is reported as OEM for MPP Solar, Axpert and others (forum, 2017) (powerforum).
- No Pakistani rebadge name appears in the open-source brand lists checked. Inference: most rebadges match an existing platform map, so platform identification matters more than brand names.

Local players

Player	What it does	Scale claimed	Role for Product 2
Enercon Systems Intl, Karachi (likely "EnerCon")	Microgrid controls on Siemens PLC; genset sync; SCADA; cloud dashboard (Enercon)	IPAK (5 gensets, 16 inverters); Lucky Energy 72 MW; markets KSA	Competitor at large microgrids; possible buyer of a driver or battery-trust module
Green Fuel Energy, Lahore (likely "Greenfuels")	EPC; sole Pakistan distributor of NeoTAQ; distributes Elum (GFE)	250+ projects, 100 MW+; supplies other EPCs	Channel tied to rivals; proves EPCs resell third-party controllers
NeoTAQ Systems, Masdar City (Pakistani team)	neoCore OEM-agnostic gateway (RS485, Modbus, OPC-UA, 4G); BESS and hybrid dispatch; cloud (NeoTAQ)	50+ PK sites; 100 MW+; KSA show 23 to 25 Nov 2026	Closest competitor; no device list or battery trust published
SkyElectric, Islamabad	Own inverter and battery; SmartFlow AI; NOC (SkyElectric)	10,000+ systems; 250+ installers	Closed ecosystem; competitor in homes and SMEs
Reon Energy	SPARK telecom EMS; REFLEX BESS (Reon)	5,500 towers; KSA licence	Telecom incumbent; phase 2
Enerlytics Lab, Karachi	Cloud monitoring; "device agnostic" with no list; no control shown (Enerlytics)	Small team; KSA distributor	Possible dashboard partner

"NeoSat Renewable" was not found. Searches returned only Neosat satellite receivers (OLX). The likeliest match is NeoTAQ, GFE's controller partner; the engineering manager should confirm.

Inference: the three names form two controller ecosystems, Enercon and GFE plus NeoTAQ. Treat both first as competitors to learn from, then as possible component buyers.

Who else is doing this

At least 14 commercial products read and control multi-brand devices, but none found sells to Pakistan's hybrid segment through local installers. SolarAssistant is closest in price and brand mix, but its shop does not ship to Pakistan, Saudi Arabia or the UAE (shop).

Player	Device library	How added	Control	Pricing (date)	Customers	PK or KSA	Link
Solar-Log Base	130 to 180 makers (page conflicts)	Vendor drivers via firmware	Feed-in rules; 100 devices	Per-function licences	260,000 plants (2017-18 headline)	None found	Solar-Log
meteocontrol blue'Log	30 to 100 devices	Vendor list	XC: P and Q control	By plant kW	C&I to utility	None found	datasheet
Eniris SmartgridOne	About 115 inverter brands	Vendor	Zero export; 600 s fallback	SMA API fees passed on	Belgium	None found	Eniris
SMA Data Manager M	50 devices; SunSpec third parties	SunSpec	P and Q control	Not found	OEM	Not found	SMA
Huawei SmartLogger3000	80 devices	Huawei	Zero export	Not found	C&I BESS bundles	KSA marketing (headline)	datasheet
Sungrow Logger1000	30 devices	Sungrow	Power control, firmware	Not found	OEM	PK distribution 2021 (headline)	Sungrow
Solarman Business	Own loggers, "third-party inverters"	ODM supplier	"Control" claimed	Quote	1.5M+ users	Sticks common (inference)	Solarman
Elum ePowerControl	"95% of the market"	Vendor	Genset load, zero export, peak shaving	Quote	PK chemical factory, 200 kWp	Via GFE	Elum
ComAp InteliNeo	Gensets via Modbus; Fronius via SunSpec	Paid SKU	PV, BESS, genset	Feature SKUs	Microgrids	Not found	ComAp
Victron Cerbo GX	4 PV brands; 7 genset brands; community drivers	Vendor and community	Zero feed-in with 2 brands	Not found	Large community	Not found	Victron
SolarAssistant	16 inverter brands, 18 BMS	Vendor	Remote settings	USD 59 software; USD 179 to 229 kits (30 Oct 2026)	South African community	No shipping	SolarAssistant

Player	Device library	How added	Control	Pricing (date)	Customers	PK or KSA	Link
gridX	55+ OEMs, 500+ devices	OEM certification	HEMS	Not found	250k+ assets	None found	gridX
Smartfox Pro 2	Modbus, RS485, S0	Vendor	Surplus control	EUR 1,130.08 incl. VAT, 10 years monitoring (30 Oct 2026)	Europe	None found	Galaxus
NeoTAQ neoCore	"OEM-agnostic"; no list	Not stated	Rule-based, BESS, genset	Not public	50+ PK sites	PK via GFE; KSA show	NeoTAQ

Business-model lessons

- Four pricing models exist: per-function licences (Solar-Log), one-off software (SolarAssistant), hardware with bundled monitoring (Smartfox), and white-label OEM deals (gridX).
- SolarAssistant tried paid updates in 2024 at USD 14 a year, then made updates free (terms). Inference: prefer an installer fleet tier over end-customer subscriptions.
- "Ready for gridX" OEM certification claims 75% fewer support tickets (gridX). Pakistani distributors of Chinese brands could play that role.
- Coverage is a moat kept by a vendor driver team (Solar-Log); a tested data-file library can grow faster.

Liability handling

- Disclaimers: Solar-Log's database says "All information without guarantee!"
- Fail-safe defaults: Eniris reverts to a preset within 600 s of link loss.
- Warranty friction: some setups need automatic updates disabled, which "manufacturers often don't approve" (Memodo).
- No vendor warranty statement on third-party Modbus writes was found.

Open-source codebases

About 40 projects cover roughly 160 inverter, meter and storage brands, but no brand list checked names a Pakistani rebadge. Stars were read from GitHub pages on 30 and 31 October 2026; last commits come from clones on 30 October 2026.

Repo	Licence	Stars	Last commit	Brands	Format	Auto-detection	Writes	Reuse verdict
ha-solarman	MIT	516	2026-05-03	Deye, Sofar, Solis, SRNE, Kstar and more (30 profiles)	YAML	Deye register 0	Yes	Schema base and data, with notices
evcc	MIT, except sponsor parts	7.5k	2026-10-30	134 meter brands	Templated YAML	Network hints; SunSpec probe	Yes, sequences	Copy template design
solax-modbus	Apache-2.0	528	2026-10-28	SolaX, Growatt, Sofar, Solis, SRNE	Python dataclasses	Serial prefix	Yes	Facts and quirks
goodwe	MIT	179	2026-03-29	GoodWe	Python classes	UDP discovery, try families	Yes	Reuse
SunGather	GPL-3.0	214	2026-10-24	Sungrow (92 codes)	YAML	Register 5000	Not reviewed	Reference only
mkaiser Sungrow	MIT	708	2026-09-06	Sungrow SH	HA YAML	Register 5000	Yes	Reuse facts
grott	None	534	2025-05-26	Growatt cloud protocol	JSON offsets	Frame header	Yes	Facts only
sunsynk	Apache-2.0	340	2026-10-24	Deye, Sunsynk	Python tables	Registers 0 to 7	Yes	Reuse
huawei-solar-lib	AGPL-3.0	48	2026-08-21	Huawei	Python dict	FC 43/14	Yes	Reference only
pysunspec2, models	Apache-2.0	74, 131	2026-10-21, 2026-08-20	Any SunSpec device	JSON	"SunS" marker	Yes	Adopt for naming
PythonProtocolGateway	Apache-2.0	65	2025-08-12	Growatt, EG4, SRNE, Pylon (48 maps)	CSV	Plausibility scoring	Partial	Reuse CSVs, scoring idea
dbus-serialbattery	MIT (archived); fork AGPL since 2024-12-06	591, 276	2026-06-25, 2026-09-29	20+ BMS brands	Python classes	Ordered driver list	Sends limits to Victron	MIT era only

Repo	Licence	Stars	Last commit	Brands	Format	Auto-detection	Writes	Reuse verdict
Battery-Emulator	GPL-3.0	3.0k	2026-10-29	57 batteries, 24 inverter protocols	C++	User selects	Emulates battery	Reference only
dbus-modbus-client	None shown	33	2026-07-10	Meters, gensets	Python handlers	Model-register probe	Not reviewed	Pattern only
OpenEMS	EPL-2.0; UI AGPL-3.0	1.6k	2026-10-30	211 bundles	Java	Configured	Yes	Data model ideas
mpp-solar	MIT	438	2026-04-12	Voltronic PI16 to PI41, JK, Daly	Command dicts	QPI	26 setters	Reuse
pysolarmanv5	MIT	180	2026-03-08	Any Solarman-logged inverter	Transport	Scan utility	Via Modbus	Reuse
foxess_modbus	MIT	333	2026-06-24	FoxESS, rebadges	Python	Model regex plus firmware	Yes	Firmware pattern
esphome-smg-ii	Apache-2.0	165	2026-10-28	SMG-II clones, PowMr, Anenji	YAML	User selects	Not reviewed	Reuse YAML

pymodbus (BSD-3-Clause, 2.8k stars) suits host-side test harnesses (pymodbus).

What the important repos teach

- **ha-solarman:** each `inverter_definitions/*.yaml` item has `rule`, `registers`, `scale`, `lookup`, `validation` and `configurable` (deye_hybrid.yaml). It ships no tests.
- **evcc:** `templates/definition/meter/sungrow-hybrid.yaml` adds aliases, firmware requirements, discovery hints and a write `sequence` (file). `cmd/detect/definitions.go` scans ports 502 and 1502, then probes SunSpec models on common unit IDs (file).
- **solax-modbus:** `const.py` defines scan groups, sleep modes and protocol-version gates (const.py).
- **goodwe:** `goodwe/__init__.py` matches serial tags, then tries each family in turn (file).
- **SunGather:** `registers-sungrow.yaml` filters registers by model and splits reads around a WiNet-S bug (file).
- **grott:** `grottdata.py` XOR-decodes Growatt frames with the mask "Growatt" (file).

- **sunsynk:** `src/sunsynk/identity.py` maps register 0 to a family; `www/docs/guide/` documents wiring and failures (file).
- **huawei-solar-lib:** `device_discovery.py` uses FC 43/14, including private object 0x87 (file).
- **pysunspec2:** `sunspec2/modbus/client.py` tries 40000, 0 and 50000, then walks the model chain (file).
- **PythonProtocolGateway:** its `analyze_protocol` routine in `classes/transports/modbus_base.py` scores every candidate map by value plausibility (file).
- **dbus-serialbattery:** caches the last good driver and runs a full scan only after 3 failures (file).
- **Battery-Emulator:** `test/` replays recorded CAN logs, a pattern worth copying (repo).
- **Victron:** `probe.py` reads one model register per handler; `scan.py` locks baud, then scans IDs (scan.py).
- **OpenEMS:** shared "Nature" channels fix signs: battery charge negative, grid import positive (SymmetricEss.java).
- **mpp-solar:** `mppsolar/protocols/pi30.py` defines Voltronic queries and setters (file).
- **pysolarmanv5:** ships `solarman-tcp-proxy` and `solarman-rtu-proxy`, which let standard Modbus tools talk through a logger and log every transaction (utilities).

Licence rules and a clean-room path

- MIT, Apache-2.0 and BSD data or code may enter a closed product if notices are kept.
- GPL and AGPL code may not; AGPL also reaches a hosted portal. Use these repos for facts only.
- EPL-2.0 is file-level: modified OpenEMS files must be published.
- No licence means all rights reserved; evcc's sponsor parts are excluded from MIT (exclusions.md).
- Sungrow's PDF forbids redistribution, so ship derived tables, not the PDF.

Clean-room steps (legal points need counsel):

1. Team A reads copyleft code and writes a neutral spec per family from vendor PDFs and captures, with no code.
2. Team B, which has not read that code, writes plugins and data files from the spec.
3. Each data file keeps a provenance log: source document, capture IDs, author, licence.
4. Bootstrap from ha-solarman and PythonProtocolGateway with a converter, then re-verify against PDFs and captures.

Methods to generalise

A ten-layer method gets as close to "detect all patterns" as the evidence allows. Templates and identification do most of the work; inference narrows the rest for a human to confirm.

The meeting's "each company has its own pattern" is not Modbus framing, which is public. It is the register map, the identity convention and the timing rules.

Layer	What it does	Evidence	Expected accuracy
0. Transport discovery	Common baud and IDs first, lock baud, then scan IDs; TCP 502, 1502, 6607, 8899	Victron two-stage scan (scan.py)	Nearly all wired devices (estimate); brute force over 6 bauds, 3 parities and 247 IDs takes 7 to 8 minutes
1. Identification	SunSpec marker, FC 43/14, model register, serial prefix, QPI, firmware read	pysunspec2, huawei-solar-lib, ha-solarman	Near-certain for templated families (estimate)
2. Templates	Data file per family, firmware variants, golden test	evcc, ha-solarman	As good as its dumps
3. Plausibility scoring	Decode candidate maps read-only and score them	PythonProtocolGateway	Unmeasured; good among sibling maps
4. Passive capture	Log the vendor dongle's polls and writes	pysolarmanv5 proxies; sunsniff	Finds live registers, not meaning
5. Type and scale guesses	Try 16 and 32-bit, both word orders, float, ASCII; anchor scale on 50 Hz, 230 V, 51.2 V	Proposed	Hypothesis

Layer	What it does	Evidence	Expected accuracy
6. Physics checks	P about V × I within 5%; power balance; energy equals integrated power; SoC trend matches battery sign	Proposed	Hypothesis: high on critical points
7. Reference correlation	Clip a known meter at the grid point and cross-correlate	Proposed	Grid register, sign and scale within minutes (estimate)
8. Whitelisted probing	Small safe change, then diff registers; never blind writes	DynPRE; SePanner	Finds writable setpoints
9. LLM drafting from PDFs	Strict JSON, lint, live validation, human sign-off	Nunes et al.	Drafting aid; never auto-enables writes

Layers 0 to 3 run on the edge; 4 to 8 run in the installer tool; 9 runs in the back office. The installer confirms 10 to 30 control-critical points before any write.

Research literature

Line of work	Result	Meaning for Product 2
Passive reverse engineering: BinaryInferno, NDSS 2023	On Modbus, precision 1.00 but recall 0.40 with priors (paper)	Finds field boundaries, not meaning
Active reverse engineering: DynPRE, NDSS 2024	Modbus F1 0.70, perfection 0.49 vs 0.09 to 0.28 passive	Probing a live device beats statistics
ICS semantics: ACSAC 2014, SePanner 2023, PVParser 2026	Process variables from traffic and multi-state diffs (2014; PVParser)	Closest analogues; none tested on inverters
Building metadata: Hong et al. 2015	Over 92% within, over 82% across buildings (arXiv)	Behaviour-based typing near 80 to 90% (estimate)

Line of work	Result	Meaning for Product 2
Active learning: Scrabble 2018	83.2% with 10 labels; 99% with 100 to 160	Human labels close the gap
LLM table extraction: Nunes 2025	GPT-4o 89.6% GriTS F1 on simple tables; 5.2% extra rows	Real manuals are harder

No paper was found that infers inverter or BMS register meaning (ICS survey 2025). Honest expectations: near-certain identification for templated families; about 80 to 90% typing for unknown devices, raised on critical points by physics checks. Claim "auto-configure" only for listed devices, and "guided onboarding in under an hour" for the rest.

Each confirmed mapping becomes a template and labelled data, so later units of the same platform need fewer labels.

Device-definition schema sketch

Block	Fields	Borrowed from
Identity	schema version, id, product aliases incl. rebadges, category	evcc `products`
Transport	kind (RTU, TCP, V5, Voltronic, CAN, Pylon RS485), baud list, parity, unit IDs, max read, barriers, turnaround, timeout	Victron, ha-solarman
Identify	probes: function code, address, type, words, match (set, regex, prefix), capture as serial or firmware	Victron `ModelRegister`
Variants	firmware or protocol ranges with overrides	foxess, solax
Points	key, function code, address, address base, type, word order, scale or scale register, unit, sign, sentinels, lookup, range, poll class, canonical name	ha-solarman, evcc, SunSpec
Writes	function code, min, max, step, persistence class, atomic group, unlock, revert register, rate limit	SunSpec, Growatt, SolaX
Commands	intents mapped to write sequences: set mode, battery power, export limit, TOU slot, release	evcc `sequence`

Block	Fields	Borrowed from
Requirements	minimum firmware, dongle, port	evcc
Tests	golden dump, expected values, source document, capture IDs, licence	Battery-Emulator

Common data model

Domain	Points	Sign convention
Grid	P per phase and total, Q, V, f, import and export energy	Import positive
PV	P per MPPT and total, energy	Production positive
Battery	P, V, I, BMS SoC, OnyxLeaf SoC with uncertainty, SoH, temperatures, cell extremes, limits, capacity	Charge negative
Load	P, energy	Consumption positive
Inverter	status, alarms, rated power, mode	None
Controls	mode, battery setpoint, export limit, TOU slots, reserve SoC, release	None
Meta	identity, firmware, template ID, quality flags, trust score	None

Names follow SunSpec; signs follow OpenEMS (ElectricityMeter). This set is the overlap of evcc, OpenEMS and SunSpec, and the physics checks can verify all of it (evcc api.go).

Trusting battery data

On LFP, 1 mV of open-circuit voltage (OCV) error can mean about 5% SoC error (Paizulamu et al. 2024). A cheap BMS's SoC is therefore a noisy guess, and the meeting's complaint need not mean bad faith.

Why OEM SoC is wrong on LFP

- Temperature: at one OCV, SoC differs by 12.89 points between 50 and 25 °C, and 17.42 points between 25 and −10 °C (same paper).
- Low-current blindness: a JK unit erred up to 50% below 1 A; 250 mA unmeasured for 7 days is 42 Ah (Victron forum).

- Field case: at cutoff, a JK BMS showed 85% and a BMV-712 shunt 57%; the 230 Ah pack was empty (same thread).
- Another case: BMS at 87% while voltage suggested about 61% (Power Forum).
- Rare resets: SoC is usually corrected only at full charge (same sources).

Decoding errors can also produce wrong battery data: untreated 0xFFFF sentinels, wrong sign or x10 scale. Test this hypothesis on the units the manager saw. Inference: hot sites that rarely reach 100% reset rarely, so drift builds for days or weeks.

Independent sensing options

Option	Accuracy	Cost (date)
Victron SmartShunt 500 A	Current 0.4%, 0.01 A resolution, offset under 20 mA (datasheet)	US$100.30, AgileOffroad, 30 Oct 2026 (listing); no Pakistani price found
Custom shunt with isolated ADC (INA228 class)	Not sourced	Not sourced; likely cheaper at volume (inference)
Clamp-on DC Hall CT	Drifts at low current (forum claim)	Not sourced; not for coulomb counting
BMS cell voltages over RS485 or CAN	Not quantified	No extra cost
Own pack voltage, 2 to 4 NTC probes, listen-only CAN	Not quantified	Low

Estimation methods

Method	Reported accuracy	Limit
Adaptive multi-model EKF (2024)	SoC RMSE 2.97% at −10 °C vs 13.05% for a UKF	Lab LFP cells
ICA for SoH (2018)	Under 2.5% error	NASA cells, not LFP
Physics-informed network for SoH (2024)	MAPE 0.87% on 387 batteries	LFP share unclear
BatteryML benchmark (2023)	Linear model error 90 cycles vs LSTM 11,911	Simple models win on small data

Recommended approach

1. Measure shunt current, pack voltage and temperature; treat BMS SoC and cell data as extra noisy inputs.
2. Run a 1-RC or 2-RC circuit model with temperature-indexed OCV and hysteresis in an EKF at 1 Hz.
3. Reset at anchors: full charge, the low-voltage knee, and long rests below 20% or above 90%. Widen uncertainty on the 30 to 80% plateau.
4. Track SoH from amp-hours between anchors, ICA peaks on slow charges, and resistance from current steps.
5. Compute a trust score from the gap between BMS SoC and the estimate; flag gaps above, say, 10 points.
6. Publish SoC with its uncertainty so Product 1 can hold a margin.

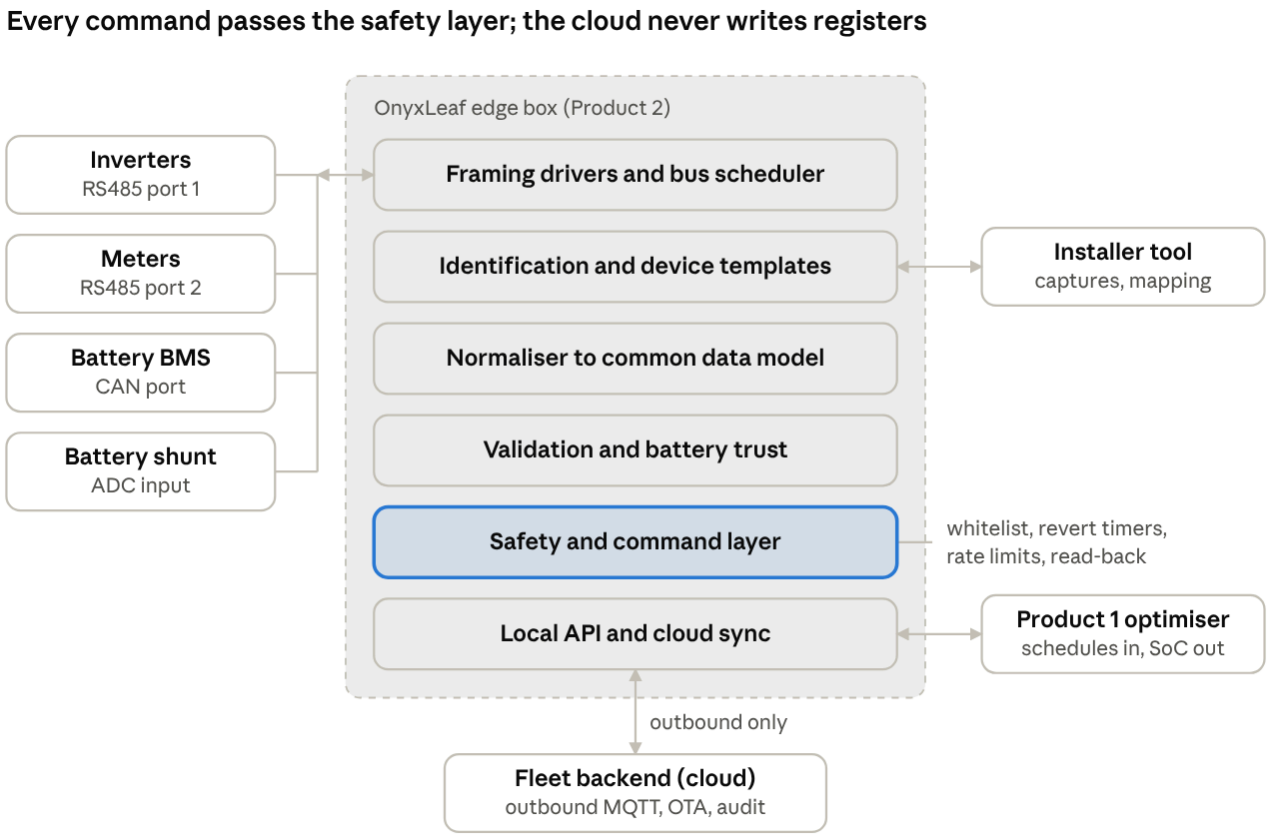
Expect about 3 to 8% field SoC error between resets; this is an inference to validate.

Datasets and limits

BatteryML unifies MATR (180 LFP cells), HUST (77 LFP) and others (BatteryML); PyBaMM can generate synthetic LFP data (PyBaMM). No public dataset covers hot-climate stationary packs read through a cheap BMS. Logging 1 Hz data from the first 20 to 50 pilot sites matters more.

Controller hardware and software architecture

A Linux module with two isolated RS485 ports, one isolated CAN port and a shunt input covers every interface found. A prototype costs an estimated US$120 to 200. Prices were retrieved on 30 October 2026 unless stated.



Product 2 controller · 6 software layers on one edge box

Telemetry flows down the stack to the local API; commands from Product 1 or the fleet enter at the API and reach a device only through the safety layer.

Hardware options

Option	Price (source)	Isolation	Fit
LilyGO T-CAN485, ESP32 with RS485 and CAN	US$11.43, Dec 2021 (CNX)	Not stated	Bridge only
Pi 5 or CM5 with Waveshare Isolated RS485 CAN HAT (B)	CM5 from US$45 at launch (Phoronix); HAT GBP 21.20, sold out (The Pi Hut)	Yes	Bench and early pilots
RevPi Connect 5 (CM5, RS485, CAN FD options)	EUR 691.39; "as low as" EUR 581.00 (Revolution Pi)	Industrial	Rugged pilots; US$800 to 1,100 with LTE (estimate)

Option	Price (source)	Isolation	Fit
Seeed reTerminal DM (CM4, screen, RS485, CAN; up to 50 °C)	US$428.75, out of stock (RobotShop)	Industrial	50 °C is tight for plant rooms (inference)
Teltonika TRB246 LTE Modbus gateway	NZ$445 ex GST (GoWiFi)	Surge only	Bridge; safety moves to cloud
Custom CM5 carrier	US$90 to 150 BOM at 1,000 units (inference)	Designed in	After pilots

Recommended build

- Compute: CM5 or CM4 on an industrial carrier; Pi 5 with the Waveshare HAT on the bench.
- Ports: 2 isolated RS485 (inverters and meters apart), 1 isolated CAN, TVS on each. Inverter ports can sit at DC-bus potential, so isolation is mandatory (inference).
- Battery: shunt ADC or VE.Direct input, pack voltage and NTC inputs.
- Power and storage: 12 to 48 V DC, RTC, supercap; industrial eMMC, read-only root, A/B partitions.
- Links: Ethernet, Wi-Fi and a PTA-approved LTE Cat 1 or Cat 4 module.
- Security and supervision: secure element for keys; MCU watchdog that can cut the write path.

Software layers

1. Framing drivers: Modbus RTU and TCP, Solarman V5, GoodWe AA55, Voltronic ASCII, Pylon RS485, CAN.
2. Bus scheduler: one queue per port; block limits, delays, atomic groups, adaptive timeouts.
3. Device templates with golden tests.
4. Identification engine (layers 0 to 3); fails closed to read-only.
5. Normaliser into the common data model, with quality flags.
6. Validation monitor: ranges, monotonic counters, physics checks, canary reads after firmware changes.
7. Battery trust estimator.
8. Safety and command layer.
9. Local API: telemetry and trusted SoC out; signed schedules and setpoints from Product 1 in.

10. Cloud sync: outbound mutual-TLS MQTT, store and forward.
11. Installer tool: survey, capture, assisted mapping, sign-off.
12. Fleet backend: registry, template distribution, OTA, audit logs.

Safety rules

1. Whitelist writable registers per template; refuse all others, and never touch grid-protection settings.
2. Tag each writable register volatile, dispatch or EEPROM; treat unknown as EEPROM.
3. Prefer registers with revert timers: SolaX remote control, Growatt 30216, Solis dispatch (1 to 1,440 min), SunSpec reversion (Solis).
4. Refresh setpoints as a heartbeat; set fallback timers explicitly, since Fronius defaults to 0, meaning never (Fronius).
5. Write on change only; cap EEPROM-class writes at, say, 100 a day per register (an engineering default).
6. Read back every write; respect atomic groups and unlock steps.
7. On a crash, stop writing and let the inverter revert; never leave forced discharge latched.
8. Log every write with who, why, and before and after values.

Cybersecurity

- SUN:DOWN (March 2025): 46 new vulnerabilities in Sungrow, Growatt and SMA (Forescout).
- A Solarman token flaw allowed any account takeover, and Deye Cloud had a hardcoded account; Solarman serves about 195 GW (Dark Reading).
- Reuters reported undocumented radios in some Chinese inverters, May 2025 (pv magazine).
- The EU restricted funding for projects with high-risk inverters from 1 May 2026 (pv magazine). An EU expert group urged outbound-only links, 8 Oct 2026 (pv magazine). CRA reporting applies from 11 Sep 2026, product rules from 11 Dec 2027 (McCann FitzGerald).
- NIST IR 8498 sets smart-inverter guidelines (NIST). SunSpec asks for signed updates, TLS 1.2+, login lockout and 90-day logs; Modbus RTU is exempt (SunSpec).

Design implications:

1. Enforce limits at the edge; the cloud sends signed schedules, never raw writes.
2. Open no inbound ports; use per-device keys and on-demand support tunnels.
3. Use secure boot and signed A/B updates; publish an SBOM per release.

4. Stagger fleet commands and require two-person approval for fleet-wide changes.
5. Let installers disable OEM dongles where OnyxLeaf takes control.

Certification

- Pakistan: devices with cellular modules need PTA type approval per module and per finished product, with accredited lab reports and GSMA TAC (PTA FAQs). Approval takes about 1 week once complete; software updates need none.
- Saudi Arabia: SABER certificates from SASO-approved bodies cover safety and wireless from 16 Feb 2025; CST (formerly CITC) sets the specs (UL). CST requires IoT type approval; foreign SIMs may stay 120 days (Eleos).
- IEC 62368-1 and IEC 61000 or CISPR 32 reports serve the PTA, SABER and CE paths; lab costs were not sourced. No rule on third-party DER controllers was found, so confirm with NEPRA, DISCOs and SERA before selling export control.

Methodology and validation

Validation runs in three stages, a 7-item bench, a capture protocol per brand and 3 to 5 pilot sites, each with numeric pass marks. All pass marks are proposed targets, not sourced figures.

Lab bench: buy first

Device	Why
Sungrow SH hybrid with WiNet-S	Named family; LAN vs dongle gap
Growatt SPH or MIN TL-XH, plus SPF	Named family; atomic writes
Deye single-phase hybrid with Solarman stick	Rebadge platform; V5 transport
Locally sold Voltronic-lineage off-grid unit	ASCII protocol; rebadges
Huawei SUN2000 with SDongle (or at a pilot site)	C&I prevalence; FC 43/14
51.2 V Pylontech-protocol pack and a JK-BMS pack, each with a SmartShunt	Battery protocols and trust tests
SDM630, DTSU666, USB-RS485, Modbus gateway, CAN adaptor, Pi 5 with HAT	Reference and tooling

Inverter prices were not researched; quote through installers.

Capture protocol per brand

1. Record nameplate, inverter and dongle firmware, connection path and wiring photos.
2. Read identity and firmware registers; store raw words.
3. Sweep documented windows read-only at the block limit; log exceptions per address.
4. Log 1 to 5 s data for 72 hours, including an outage, beside a reference meter and shunt.
5. Capture the vendor dongle's polls and app-made writes where safe.
6. Test each whitelisted write, then power-cycle to learn if it persists.
7. Save a golden dump with expected values and provenance.

Test plan and acceptance criteria

Test	Pass mark (proposed)
Decode correctness vs vendor app	100% match in scale and sign on control-critical points
Read accuracy vs reference meter, 7 days	Within 3% of reading, or 30 W at low power
Identification	100% on bench; no wrong-template write in the field
Write safety	Zero off-whitelist writes; 100% read-back match; EEPROM writes at most 100 a day per register
Fallback after controller power or link loss	Inverter back in own mode within 5 minutes
Battery trust over 30 days	OnyxLeaf SoC within 5 points at anchors; BMS gaps over 10 points flagged
30-day soak	At least 99.5% data completeness at 5 s; no unrecovered lockups; no safety incidents
Onboarding	Listed device in 15 minutes; unlisted device's critical points in 60 minutes

Pilot sites

Run 3 to 5 sites through installers, each with two or more brands and a battery, and one with a genset. Sign terms on liability, data rights and write scope first.

Each site runs read-only for two weeks, then with Product 2 commanding modes and export limits. Product 1′s saving test can share the sites; a Saudi pilot waits for SABER approval and a local partner.

Risks and open questions

Twelve risks matter most; the top three are over-promising coverage, silent map changes after firmware updates, and writes that damage equipment.

Risk	Why it matters	Mitigation
"Works with all systems"	Trust falls at the first unsupported unit	Tiered list: tested, community, read-only
Firmware changes maps	Deye x10 scale, slave-ID reset	Firmware-keyed variants; canary reads
Writes damage devices or warranties	EEPROM wear; no OEM statement found	Whitelist, rate limits; written OEM confirmation
Bus contention	Single master; dongles block serial	Topology survey; read through inverter; proxy
Strong local rivals	NeoTAQ and Elum sold via GFE	Chinese hybrid coverage, battery trust, price
OEM counter-moves	Cloud-only or encrypted dongles	Wired RS485 stays plain Modbus
Fleet cyber risk	One box commands many sites	Local-first, signed schedules, staggered commands
Licence contamination	Copyleft code in closed firmware	Clean room and provenance logs
Unknown brand mix	Wrong first templates	Census of 30+ sites first
Shunt adds cost	Higher box price	Optional shunt; BMS cell data as fallback
Regulation	PTA and SABER per product; export rules unclear	Pre-approved modules; ask NEPRA and SERA
Weak Saudi case	Flat tariffs; 5 halalas export	Diesel-heavy sites via a partner

Open questions for the engineering manager

1. Who is "NeoSat Renewable"? Is it NeoTAQ Systems, Green Fuel Energy's controller partner?
2. Are "EnerCon" and "Greenfuels" Enercon Systems (Karachi) and Green Fuel Energy (Lahore)? Is each a reseller, pilot host, supplier or competitor?
3. Which "wrong battery data" case did you see: brand, BMS, fault type and how it was found?
4. Who said "buy inverter of my company", and in what deal?
5. Which installers can open 30 census sites and 3 to 5 pilot sites?
6. Will OnyxLeaf carry hardware stock and certification, or use a partner's gateway?
7. Is genset start and stop needed in release 1?
8. Should release 1 control devices from day one, or start read-only?
9. Which Saudi partner and site types are realistic?
10. Is Enerlytics Lab a dashboard partner or a competitor?
11. What team and budget exist for the first 6 months?

Work packages

Twelve work packages take Product 2 from census to field acceptance, at about 77 person-weeks in total. Effort figures are rough estimates.

ID	Name	Goal	Inputs	Deliverable	Acceptance criterion	Depends on	Effort (person-weeks)
WP1	Device census	Learn the real device mix	Installer contacts	Site census with models, firmware, dongles, ports; ranked families	30+ sites; 80% of devices identified to firmware	None	4
WP2	Bench and captures	Golden dumps for first families	WP1 ranking	Running bench; dumps per capture protocol	6 families, each with one firmware variant	WP1	6
WP3	Schema, model, test harness	Make a brand a data file plus a test	This report; SunSpec models	Schema, linter, test runner, data model	CI passes on 3 converted open-source profiles	None	4
WP4	Transport and scheduler	Talk safely to all first-wave devices	Vendor PDFs	Framing drivers; per-port scheduler	72 hours on bench with no lockups	WP3	6

ID	Name	Goal	Inputs	Deliverable	Acceptance criterion	Depends on	Effort (person-weeks)
WP5	Template library v1	Cover first families, clean room	WP2 dumps; PDFs; permissive data	Templates with provenance and tests	All golden tests pass; 100% decode match	WP2, WP4	8
WP6	Identification engine	Pick templates without the installer	WP5	Discovery, identification, scoring, fail-closed logic	100% bench ID; wrong ID never enables writes	WP5	4
WP7	Safety and command layer	Safe vendor writes from abstract commands	Template write fields	Command API, whitelist, heartbeat, limits, audit log	Zero off-whitelist writes; fallback within 5 minutes	WP5	6
WP8	Battery trust module	Independent SoC, SoH, trust score	Shunt; BMS data	EKF, anchors, SoH, trust score	SoC within 5 points at anchors over 30 days	WP4	8
WP9	Edge hardware and security	A certifiable box	Hardware and cyber sections	5 pilot units; secure boot, mutual TLS, signed OTA; PTA and SABER plan	Units built; checklist passed; lab quotes in hand	None	8
WP10	Installer tool	Commission and map devices	WP6; layers 4 to 9	Survey, capture, physics checks, PDF drafting, sign-off	15 minutes listed; 60 minutes unlisted	WP6, WP7	8
WP11	Local API and sync	Serve Product 1 and the fleet	Data model	Local API, MQTT sync, registry	Product 1 reads SoC and sends a schedule end to end	WP7, WP8, WP9	5
WP12	Field pilots	Prove reliability on real sites	All above; pilot terms	Pilot report vs acceptance table	3 sites pass all marks over 30 days	WP10, WP11	10

Order: WP1, WP3 and WP9 start together; WP2 and WP4 follow; WP5 gates WP6, WP7 and WP8; WP10 and WP11 come next; WP12 runs last. WP11 is the hand-off point to Product 1.

Sources

Protocols and vendor maps

- Sungrow hybrid protocol V1.1.9; string protocol V1.1.53; WiNet-S datasheet; DTSU666 protocol
- Growatt Modbus RTU V1.24; grott registers
- huawei_solar wiki; kellerza wiring, fault-finding, mbusd
- solax-modbus baud rates, adaptors, SolaX FAQ, EEPROM note; Sungrow FAQ (mkaiser)
- pysolarmanv5 protocol; mbmd SDM; Victron attributes; PI30 protocol
- SMA Modbus TI; Fronius Modbus manual; Solis dispatch discussion

Standards

- SunSpec models; IEEE 1547-2018 slides; QualityLogic 1547.1
- Pylontech LV CAN V1.3; Sunsynk battery list v14

Open-source codebases

- ha-solarman; evcc; solax-modbus; goodwe; SunGather; mkaiser Sungrow; grott; sunsynk; huawei-solar-lib; deye-inverter-mqtt
- pysunspec2; PythonProtocolGateway; dbus-serialbattery; Battery-Emulator; dbus-modbus-client; OpenEMS; mpp-solar; pysolarmanv5; foxess_modbus; esphome-smg-ii; python-pylontech; pymodbus

Detection and inference research

- BinaryInferno; DynPRE; Hadžiosmanović et al. 2014; SePanner; PVParser; ICS RE survey
- Hong et al. 2015; Scrabble; LLM table extraction; sunsniff

Battery state

- LFP SoC estimation (arXiv 2410.23646); Victron forum JK vs BMV; Power Forum SoC thread
- SmartShunt datasheet and price; PINN SoH; ICA SoH; BatteryML paper and repo; PyBaMM

Hardware, cybersecurity and certification

- RevPi Connect 5; reTerminal DM; Waveshare isolated HAT; LilyGO T-CAN485; Teltonika TRB246; CM5 launch

- Forescout SUN:DOWN; Dark Reading; pv magazine on Reuters; EU funding restriction; EU expert group; NIST IR 8498; SunSpec cyber requirements; CRA reporting date
- PTA type approval FAQs; UL on Saudi wireless rules; Saudi IoT regulation

Commercial players

- Solar-Log; meteocontrol; Eniris; SMA EDMM-20; Huawei SmartLogger3000; Sungrow Logger1000; Sungrow EMS300CP; Solarman Business
- Elum; ComAp; Victron Cerbo GX; SolarAssistant, shop, terms; gridX; Smartfox; Memodo EMS 2024

Pakistan players and brand mix

- NeoTAQ and case study; Green Fuel Energy, comparison blog, solar-diesel controller; Enercon microgrid controls; SkyElectric; Reon; Enerlytics Lab; OLX Neosat
- Premier Energy; Nizam; Fronus; Inverex; SolaX and Fronus; Voltronic rebrands; SolarCitizen prices

Demand context

- Dawn on net billing; NEPRA SRO 1501(I)/2026; IESCO tariff guide; ESS News imports; PSA imports; Huawei and Jazz (DCD)
- Diesel price, 30 Oct 2026; Enpower genset costs; Haala Energy on Saudi rules